

# Question Bank on Linear Combination, Linear Dependence & Independence of Vectors, and Span.

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## Part A – Conceptual / Definitions

### 1. Define linear combination of vectors over a field $F$ .

- **Solution:** A linear combination of vectors  $v_1, v_2, \dots, v_n$  in a vector space  $V$  over a field  $F$  is any vector of the form  $c_1*v_1 + c_2*v_2 + \dots + c_n*v_n$  where  $c_1, c_2, \dots, c_n$  are scalars in  $F$ .

### 2. What does it mean if a vector $v$ is a linear combination of $v_1, v_2, \dots, v_n$ ?

- **Solution:** It means there exist scalars  $c_1, c_2, \dots, c_n$  in  $F$  such that  $v = c_1*v_1 + c_2*v_2 + \dots + c_n*v_n$ .

### 3. Define linear dependence and linear independence in a vector space over a field $F$ .

- **Solution:** Vectors are linearly dependent if there exist scalars (not all zero)  $c_1, c_2, \dots, c_n$  such that  $c_1*v_1 + c_2*v_2 + \dots + c_n*v_n = 0$ . (Null Vector).
- They are linearly independent if the only scalars satisfying that equation are all zero.

### 4. State the condition for a set of vectors $\{v_1, \dots, v_n\}$ in $R^n$ to be linearly dependent.

- **Solution:** The set is linearly dependent if at least one vector can be expressed as a linear combination of the others (equivalently the column-rank  $<$  number of vectors).

### 5. Define linear span of a set of vectors.

- **Solution:** The span of  $\{v_1, \dots, v_n\}$  is the set of all linear combinations of these vectors:  $\text{span}\{v_1, \dots, v_n\} = \{c_1*v_1 + \dots + c_n*v_n \mid c_i \in F\}$ . It is a subspace of  $V$ .

### 6. Can the zero vector be a part of a linearly independent set? Why or why not?

- **Solution:** No. If  $0$  is in the set, then  $1*0 + 0*\text{others} = 0$  gives a nontrivial relation, so the set is dependent.

### 7. What is the criterion of linear independence using the rank of a matrix?

- **Solution:** Place vectors as columns of a matrix. They are independent iff the matrix rank equals the number of vectors (no nontrivial linear relation).

### 8. Define the vector space of polynomials of degree $\leq 3$ over $R$ .

- **Solution:**  $P_3(R) = \{a_0 + a_1 x + a_2 x^2 + a_3 x^3 \mid a_0, a_1, a_2, a_3 \in R\}$ .

**9. Write an example of a vector space consisting of  $2 \times 2$  matrices over  $\mathbb{R}$ .**

- **Solution:**  $M_2(\mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}$  with usual matrix addition and scalar multiplication.

## **Part B – Short Answer / Analytical**

**10. Check whether the vectors  $(1,2,3)$ ,  $(2,4,6)$ ,  $(3,6,9)$  are linearly independent in  $\mathbb{R}^3$ .**

- **Solution:** Note  $(2,4,6) = 2 \cdot (1,2,3)$  and  $(3,6,9) = 3 \cdot (1,2,3)$ . Thus nontrivial relations exist; the set is linearly dependent.

**11. Determine whether the complex vectors  $(1+i,2)$ ,  $(2+i,4+2i)$  are linearly independent in  $\mathbb{C}^2$ .**

- **Solution:** For two vectors in  $\mathbb{C}^2$  to be dependent, one must be a complex scalar multiple of the other.  
Check  $\lambda$  such that  $(2+i,4+2i) = \lambda(1+i,2)$ . From second component  $\lambda = (4+2i)/2 = 2+i$ .  
Check first:  $\lambda(1+i) = (2+i)(1+i) = (2+i) + (2i-1) = (1+3i) \neq 2+i$ , so not a scalar multiple.  
Therefore, they are linearly independent.

**12. For what value of  $k$  are the vectors  $(1, 2, k)$ ,  $(2,3,4)$ ,  $(3,4,5)$  linearly dependent?**

- **Solution:** Form determinant of matrix with these as columns (or rows). Determinant simplifies to  $(3-k)$ . Setting determinant=0 gives  $k=3$ . Thus dependent when  $k=3$ .

**13. Determine whether matrices  $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ,  $C = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  are linearly independent over  $\mathbb{R}$ .**

- **Solution:** Since  $C = A + B$ . Hence there is a nontrivial relation  $C - B - A = 0$  (Null Matrix), so dependent.

**14. Determine whether polynomials  $p_1(x)=1+x$ ,  $p_2(x)=1-x$ ,  $p_3(x)=x^2$  are linearly independent in  $P_2(\mathbb{R})$ .**

- **Solution:** Write coefficients:  $p_1 = [1,1,0]$ ,  $p_2 = [1,-1,0]$ ,  $p_3 = [0,0,1]$ . The coefficient matrix is block diagonal with determinant  $-1 \neq 0$ . Hence independent.

**15. Show that  $(1,0,0)$ ,  $(0,1,0)$ ,  $(0,0,1)$  form a linearly independent set in  $\mathbb{R}^3$ .**

- **Solution:** A linear combination  $c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = 0$  give  $(c_1, c_2, c_3) = 0$ . So only trivial solution; independent.

**16. If  $v_1 = (1, i)$ ,  $v_2 = (i, -1)$ , test for linear independence in  $\mathbb{C}^2$ .**

- **Solution:** Check if  $v_2 = \lambda v_1$  for some  $\lambda \in \mathbb{C}$ . From  $v_2 = i v_1$   
 $i \cdot (1, i) = (i, i^2) = (i, -1) = v_2$ . So  $v_2 = i v_1$ , therefore dependent.

**17. Determine whether (1,1,0), (0,1,1), (1,0,1) are linearly independent.**

- **Solution:** Form matrix  $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ ; determinant = 2  $\neq$  0, so independent.

**18. Find the relation among (1, -2, 1), (2, -3, 0), (3, -4, -1).**

- **Solution:** Solve  $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ . One solution is  $c_1=1, c_2=-2, c_3=1$ , i.e.,  $v_1 - 2v_2 + v_3 = 0$ . Hence dependent.

**19. 10. Show that if one vector is a scalar multiple of another, they are linearly dependent.**

- **Solution:** If  $v_2 = \alpha v_1$ , then  $v_2 - \alpha v_1 = 0$  gives a nontrivial linear relation; hence dependent.

### **Part C – Numerical / Matrix-Based Problems**

**20. Check whether the vectors (1,2,3,4), (2,3,4,5), (3,4,5,6), (4,5,6,7) are linearly dependent using matrix rank.**

- **Solution:** Form 4x4 matrix with these columns/rows. Row operations or observing linear recurrence show rank < 4 (in fact rank=2). Therefore dependent.

**21. Determine whether (1,2,3), (2,5,8), (3,8,13) are linearly independent.**

- **Solution:** Compute determinant of matrix with these as columns: determinant = 0 (columns follow a linear relation). Hence dependent.

**22. For what values of a and b are (1, 2, 3), (a, b, 6), (4, 5, 6) linearly dependent?**

- **Solution:** Set determinant = 0: solving gives condition  $a - 2b + 6 = 0$ , i.e.,  $a = 2b - 6$ . Any (a,b) satisfying this yields dependence.

**23. Check linear dependence of matrices  $M_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ,  $M_2 = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$ ,  $M_3 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  over  $\mathbb{R}$ .**

- **Solution:**  $M_2 = 2 M_1$ , so  $M_1$  and  $M_2$  are dependent; including  $M_3$  gives dependence as well.

**24. Find whether  $p_1(x)=1+x$ ,  $p_2(x)=x+x^2$ ,  $p_3(x)=1+x+x^2$  are linearly independent in  $P_3(\mathbb{R})$  (coefficient matrix method).**

**Solution:** Coefficient matrix  $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$  has determinant 1  $\neq$  0, so independent.

**25. Determine if  $\{(1,2i), (2,4i)\}$  are linearly dependent over  $\mathbb{C}$ .**

- **Solution:**  $(2,4i) = 2*(1, 2i) \Rightarrow$  dependent.

**26. Check whether  $(1, 0, -1, 2)$ ,  $(2, 1, 0, 3)$ ,  $(3, 1, -1, 5)$  are linearly independent in  $\mathbb{R}^4$ .**

- **Solution:** Solve  $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ . Matrix Method yields only trivial solution  $\Rightarrow$  independent.

**27. Show that any set of four vectors in  $\mathbb{R}^3$  is linearly dependent.**

- **Solution:** Dimension of  $\mathbb{R}^3$  is 3. Any set with more than 3 vectors must be dependent

**28. Find the rank and dependence of  $(1, 1, 1)$ ,  $(1, 2, 3)$ ,  $(1, 3, 6)$ .**

- **Solution:** Determinant of matrix with these as columns  $= 1 \neq 0$  so rank = 3 and the vectors are independent.

**29. Verify that  $\{1, x, x^2\}$  forms a linearly independent set in  $P_2(\mathbb{R})$ .**

- **Solution:** Let  $c_1 + c_2 x + c_3 x^2 = 0$  for all  $x$ .  
Equating coefficients gives  $c_1 = c_2 = c_3 = 0$ . Hence independent.

## **Part D – Applications and Higher-Order Thinking**

**30. Explain how linear combination is used in solving systems of linear equations using Gaussian elimination.**

- **Solution:** Writing  $Ax = b$  shows  $b$  as a linear combination of  $A$ 's columns with coefficients  $x$ . Gaussian elimination finds these coefficients by systematic elimination of variables (row operations).

**31. Discuss the role of linear independence in determining unique solutions to  $AX=B$ .**

- **Solution:** If columns of  $A$  are linearly independent and  $A$  is square (full rank),  $A$  is invertible, leading to a unique solution  $x = A^{-1}B$ . If dependent, solutions may be infinite or nonexistent.

**32. In signal processing, signals  $s_1$ ,  $s_2$ ,  $s_3$  are linearly dependent. Explain what it implies physically.**

- **Solution:** Dependency means one signal is a linear combination of others; it implies redundancy, reduced information content, and possible dimensionality reduction for compression.

**33. Let  $f_1(t)=e^t$ ,  $f_2(t)=e^{2t}$ ,  $f_3(t)=e^{3t}$ . Prove they are linearly independent.**

- **Solution:** Assume  $c_1 e^t + c_2 e^{2t} + c_3 e^{3t} = 0$  for all  $t$ . Evaluate at  $t=0$  and derivatives to obtain a linear system with only trivial solution  $c_1 = c_2 = c_3 = 0$ .  
Alternatively,  
Wronskian  $\neq 0$ , so independent.

**34. Show that vectors representing the forces on a rigid body are linearly dependent iff they are coplanar.**

- **Solution:** For three forces, coplanarity means one lies in span of other two, producing a nontrivial relation; conversely, dependence implies all lie in some plane (span of two).

**35. Explain with an example how span helps in constructing basis vectors in  $\mathbb{R}^3$ .**

- **Solution:** Two nonparallel vectors span a plane; adding a third vector not in that plane gives three independent vectors that span  $\mathbb{R}^3$ , hence form a basis (example:  $(1,0,0)$ ,  $(0,1,0)$ ,  $(0,0,1)$ ).

**36. Determine if polynomials 1,  $(x+1)$ ,  $(x+1)^2$  are linearly independent in  $P_2(\mathbb{R})$ .**

- **Solution:** Expand:  $(x+1) = 1 + x$ ,  $(x+1)^2 = 1 + 2x + x^2$ .  
Coefficient matrix is upper-triangular with nonzero diagonal entries; determinant  $\neq 0$   
 $\Rightarrow$  independent.

## Part E – Mixed Problems

**37. If  $v_1 = (1,1,1)$ ,  $v_2 = (1, -1,1)$ ,  $v_3 = (1,2,1)$ , determine if  $v_3$  can be expressed as linear combination of  $v_1$  and  $v_2$ .**

- **Solution:** Solve  $a v_1 + b v_2 = v_3 \Rightarrow$  system  $a+b=1$ ,  $a-b=2 \Rightarrow a=3/2$ ,  $b=-1/2$ .  
So  $v_3 = 3/2 v_1 - 1/2 v_2$ .

**38. Prove if  $A$ ,  $B$ ,  $C$  are  $n \times n$  matrices and  $A+2B-3C=0$  then they are linearly dependent.**

- **Solution:** The coefficients  $(1, 2, -3)$  are not all zero and satisfy the linear relation; hence dependent.

**39. Find constants  $a$ ,  $b$ ,  $c$  such that  $(1, i, 0) = a(1,1,1) + b(1, -1,0) + c(0,1, i)$ .**

- **Solution:** Solve system componentwise.  
One solution:  $a = (3+i)/5$ ,  $b = (2-i)/5$ ,  $c = (-1+3i)/5$  (students may verify by substitution).

**40. Let  $V$  be continuous functions on  $[0,1]$ . Show  $\{1, \sin x, \cos x\}$  is linearly independent.**

- **Solution:** Suppose  $A + B \sin x + C \cos x = 0$  for all  $x$  in interval. Evaluate and differentiate at  $x=0$  to obtain  $A=B=C=0$ . Thus independent.

**41. Verify  $\{(1,0,0,0), (1,1,0,0), (1,1,1,0), (1,1,1,1)\}$  is linearly independent in  $\mathbb{R}^4$ .**

- **Solution:** Let  $c_1, \dots, c_4$  combination  $= 0$ . Looking at last component gives  $c_4=0$ ; third gives  $c_3=0$ ; second gives  $c_2=0$ ; first gives  $c_1=0$ . Hence all zero, independent.